Kun Huang

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Academic Appointments

Texas A&M University — Texas, US, 77840
Visiting Assistant Professor — 7/2026 - now
Postdoctoral Research Associate, Advisors: Huiyan Sang and Xiyu Peng — 9/2023 - 7/2026

Education

Tsinghua University — Beijing, China, 100084
Ph.D. in Statistics, Advisor: Lijian Yang — 8/2018 - 7/2023
B.E. in Mechanical Engineering, Minor in Statistics — 8/2014 - 7/2018

Research Interests

- Theory and methodology: Bayesian spatial machine learning; statistical learning for heterogeneous and structured data; large-scale and high-dimensional statistics; spatial causal inference; functional data analysis; spatio-temporal point processes
- Applications: statistics in spatial transcriptomics, transportation, environmental science, biomedical sciences and the emerging AI field

Submitted Papers

† co-first author; § co-corresponding author

1. **Huang, K.**, Peng, X., Sang, H. and Lu, L. (2026+) Consistent Bayesian local spatial feature selection with application to spatial multimodal omics. *Journal of the American Statistical Association* (Theory and Methods), **Major Revision**. arXiv:2605.30658. https://arxiv.org/abs/2605.30658

2. **Huang, K.**, Yang, S., Li, Y. and Guan, Y. (2026+) Interference-aware causal estimation for marked spatio-temporal point processes: assessing the effects of California wildfires on $PM_{2.5}$, *Biometrics*, **Major Revision**

3. **Huang, K.**, Peng, X. and Sang, H. (2026+) A consistent Bayesian adaptive quotient graph framework for graph-informed random partition models. *Annals of Statistics*, **Under Review**

Publications

† co-first author; § co-corresponding author

Methodology and Theory

1. **Huang, K.** and Sang, H. (2026) Consistent Bayesian spatial domain partitioning using predictive spanning tree methods. ***Annals of Statistics***, 54, 2432-2454. (Selected for presentation at the NeurIPS 2026 Journal Track)

2. **Huang, K.**, Chen, X., Guan, Y. and Li, Y. (2025) Partially linear single-index models and functional principal component analysis of spatially and temporally indexed point processes. ***Journal of Business and Economic Statistics***, 43, 1077-1091

3. Zheng, S., **Huang, K.** and Yang, L. (2025) Inference for dependent error functional data: covariance function. ***Electronic Journal of Statistics***, 19, 5216-5248

4. Chen, X., **Huang, K.**, Wu, W. and Jiang, H. (2025) Detecting multiple change points by exploiting their spatio-temporal correlations: A Bayesian hierarchical approach. ***INFORMS Journal on Data Science***, 4, 133-153

5. **Huang, K.**, Zheng, S. and Yang, L. (2022) Inference for dependent error functional data with application to event related potentials. ***TEST***, 31, 1100–1120

6. Chen, X., **Huang, K.** and Jiang, H. (2022) Detecting changes in the spatiotemporal pattern of bike sharing: a change-point topic model. ***IEEE Transactions on Intelligent Transportation Systems***, 23, 18361–18377. [JCR:Q1, IF:9.1]

Applications

1. Shi, S.†, Chen, W.†, Jiang, Y., Chen, K., Liao, Y.§ and **Huang, K.**§ (2023) A more accurate method to estimate muscle mass: a new estimation equation. ***Journal of Cachexia, Sarcopenia and Muscle***, 14, 1753–1761 [JCR:Q1, IF:9.9]

2. Chen, W.†, Shi, S.†, Jiang, Y.†, Chen, K., Liao, Y., Huang, R.§ and **Huang, K.**§ (2022) Association of sarcopenia with ideal cardiovascular health metrics among US adults: a cross-sectional study of NHANES data from 2011 to 2018. ***BMJ Open***, 12, e061789.

3. Shi, S.†, Jiang, Y.†, Chen, W.†, Chen, K., Liao, Y.§, and **Huang, K.**§ (2022) Diagnostic and prognostic value of the Creatinine/Cystatin C ratio for low muscle mass evaluation among US adults. ***Frontiers in Nutrition***, 9. [JCR:Q1, IF:5.5]

4. Chen, W.†, Shi, S.†, Jiang, Y.†, Chen, L., Liao, Y.§, Chen, K.§, and **Huang, K.**§ (2022) Association between riboflavin intake and telomere length: a cross-sectional study from NHANES 1999-2002. ***Frontiers in Nutrition***, 744397. [JCR:Q1, IF:5.5]

5. Hu, C., Zhu, K., **Huang, K.**, Yu, B., Jiang, W., Peng K. and Wang F. (2022) Using natural intervention to promote subjective well-being of essential workers during public-health crises: A Study during COVID-19 pandemic. ***Journal of Environmental Psychology***, 79, 101745. [JCR:Q1, IF:7.3]

6. **Huang, K.**†, Chen, D.†, Wang, F.§ and Yang, L.§ (2021) Prediction of dispositional dialectical thinking from resting-state Electroencephalography. ***Brain and Behavior***, 11, e2327.

7. Zhang, Y., Wang, C., Wu, F., **Huang, K.**, Yang, L. and Ji, L. (2020) Prediction of working memory ability based on EEG by functional data analysis. ***Journal of Neuroscience Methods***, 333, 108552.

Working Papers

† co-first author; § co-corresponding author

1. Ghosh, A., Sang, H. and **Huang, K.** (2026+) Consistent spatial clustering: a general framework

Patents

1. Yang, Li., Zhang, Y. and **Huang, K.** (2021) Physiological state prediction method, computer device and storage medium. 202110180615.7[P]. 2022-11-29

2. **Huang, K.** and Yang, L. (2021) Comparison method and system for event-related potential signals. 202111011898.9[P]. 2021-08-31

Scholarships and Awards

- NSF Travel Award, 2025 ICSA Applied Statistics Symposium 2025
- First Class Scholarship for Comprehensive Performance, Tsinghua University 2022
- Second Class Scholarship for Comprehensive Performance, Tsinghua University 2021
- Peking-Tsinghua Joint Statistics Colloquium Excellent Poster Award 2021

Conference Presentations

- Consistent Bayesian local spatial feature selection with application to spatial multimodal omics August 2026
  ***Contributed Session Speaker**, Joint Statistical Meetings, Boston, MA, USA*
- Consistent Bayesian spatial domain partitioning using predictive spanning tree methods June 2025
  ***Invited Session Speaker**, ICSA Applied Statistics Symposium, Storrs, CT, USA*
- Spatially adaptive Bayesian function estimation with basis selection on complex constrained domain August 2024
  ***Invited Session Speaker**, Joint Statistical Meetings, Portland, OR, USA*
- Inference for dependent error functional data with application to event related potentials August 2022
  ***Contributed Session Speaker**, International Conference on Computational Statistics (Online), Bologna, Italy*
- Inference for dependent error functional data with application to event related potentials July 2022
  ***Invited Session Speaker**, International Chinese Statistical Association China Conference (Online), Xi'an, China*
- Prediction of dispositional dialectical thinking from resting-state electroencephalography August 2021
  ***Topic Contributed Session Speaker**, Joint Statistical Meetings (Online), Seattle, WA, USA*

Teaching Experience

- **Instructor** at Texas A&M University Spring 2027
  Statistics 641: Methods of Stat I, *graduate course*
- **Instructor** at Texas A&M University Fall 2024, Fall 2026
  Statistics 302: Statistical methods, undergraduate course
- **Teaching Assistant** at Tsinghua University Fall 2019

Computational statistics, undergraduate course

- **Teaching Assistant** at Tsinghua University Spring 2019

  Work for the teaching committee of the statistical center. Help to design and prepare materials for the undergraduate course

- **Teaching Assistant** at Tsinghua University Fall 2018

  Computational statistics, undergraduate course

Academic Mentoring

- *Arpan Ghosh* Texas A&M University

Academic services

- **Reviewer**

  Journal of the American Statistical Association, Journal of Computational and Graphical Statistics, Journal of Multivariate Analysis, Statistica Sinica, Stat, Environmental Research, F1000Research